Linear & Quadratic Discriminant Analysis

Classification by probability: Bayes' rule, Gaussian class models, and the geometry of decision boundaries

1 · The classification problem, and two philosophies

Given n samples (x_i, y_i), $x \in \mathbb{R}^p$, $y \in \{1, \dots, K\}$, we want a rule $\hat{y}(x)$ that minimises errors on *new* data. Classes overlap, so zero error is impossible; the floor is the **Bayes error**.

- **Generative** models learn $P(x|y)$ and $P(y)$, then invert with Bayes — a full density per class. Strong assumptions buy statistical efficiency on small n. (LDA, QDA, naive Bayes, HMMs.)
- **Discriminative** models learn $P(y|x)$ or just the boundary — fewer assumptions, often lower error when n is large. (Logistic regression, SVMs, neural nets.)

> Ng & Jordan (2002): generative classifiers reach their asymptote with fewer samples; discriminative ones reach a lower asymptote. Small data → LDA earns its keep.

2 · Bayes' theorem as a classifier

$$P(y = k \mid x) = \frac{\pi_k f_k(x)}{\sum_j \pi_j f_j(x)}, \qquad \hat{y}(x) = \arg\max_k \pi_k f_k(x)$$

Posterior = prior × likelihood, renormalised. $\pi_k = P(y = k)$ is the class frequency; $f_k(x) = P(x|y = k)$ is how class k generates data. The denominator is shared, so only the numerator matters. With the true densities this rule has the lowest possible error.

3 · The modelling choice: Gaussian classes

$$f_k(x) = (2\pi)^{-p/2} |\Sigma_k|^{-1/2} \exp\left(-\tfrac{1}{2}(x - \mu_k)^\top \Sigma_k^{-1} (x - \mu_k)\right)$$

A multivariate normal: mean vector μ_k, covariance matrix Σ_k. The exponent is the **Mahalanobis distance** — distance in covariance units; contours are ellipsoids whose axes are eigenvectors of Σ. Why Gaussian? CLT for measurement noise, closure under marginals/conditioning, and the decisive fact: **log f is a quadratic in x**, so comparing classes is algebra, not integration.

4 · LDA: one covariance to share

Assumption: $x \mid y = k \sim \mathcal{N}(\mu_k, \Sigma)$ — the same Σ for every class: equal shape, equal orientation, clouds only shifted.

$$\log \frac{P(y = k|x)}{P(y = l|x)} = \log \frac{\pi_k}{\pi_l} + x^\top \Sigma^{-1}(\mu_k - \mu_l) - \tfrac{1}{2}(\mu_k + \mu_l)^\top \Sigma^{-1}(\mu_k - \mu_l)$$

The two-line derivation of linearity: in the log-posterior ratio, the quadratic term $x^\top \Sigma^{-1} x$ appears on both sides with the same Σ — it cancels. What remains is affine in x, so every pairwise boundary is a hyperplane. Keep per-class Σ_k instead and the quadratic survives → QDA.

$$\delta_k(x) = x^\top \Sigma^{-1}\mu_k - \tfrac{1}{2}\mu_k^\top \Sigma^{-1}\mu_k + \log \pi_k$$

The linear discriminant: classify to argmax. Three parts: a **match** term (how strongly x points toward class k), a **mean penalty** (normalises classes far from the origin), a **prior bonus** (frequent classes get a head start). Sanity check: $\Sigma = \sigma^2 I$ and equal priors → "assign to the nearest class mean", properly scaled.

$$\hat{\pi}_k = \frac{n_k}{n}, \quad \hat{\mu}_k = \frac{1}{n_k}\sum_{y_i=k} x_i, \quad \hat{\Sigma} = \frac{1}{n-K}\sum_k \sum_{y_i=k} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^\top$$

Fitting = three sample statistics: count, average, pool. Closed form, one pass, $O(np^2)$ — nothing to tune, nothing to diverge. (Contrast with RL, which iterates a stochastic update toward a fixed point.)

Worked example in one dimension

Two classes, shared σ. The boundary solves $\delta_1(x) = \delta_2(x)$:

$$x^* = \frac{\mu_1 + \mu_2}{2} + \frac{\sigma^2}{\mu_2 - \mu_1} \ln \frac{\pi_1}{\pi_2}$$

Midpoint, shifted by the priors. With $\mu_1 = 0$, $\mu_2 = 3$, $\sigma = 1.3$, equal priors: $x^* = 1.5$ (the midpoint). Tilt the priors to $\pi_1 = 0.7$: shift = $(1.69/3)\cdot\ln(0.7/0.3) \approx 0.48$, so $x^* \approx 1.98$ — the common class annexes territory, exactly the log π term at work. Exercise: $\mu_1 = 1$, $\mu_2 = 5$, $\sigma = 2$ → (a) equal priors: $x^* = 3$; (b) $\pi_1 = 0.7$: $x^* = 3 + (4/4)\cdot\ln(7/3) \approx 3.85$.

5 · QDA: every class its own covariance

$$\delta_k(x) = -\tfrac{1}{2}\log|\Sigma_k| - \tfrac{1}{2}(x-\mu_k)^\top \Sigma_k^{-1}(x-\mu_k) + \log \pi_k$$

x's quadratic term no longer cancels. New term: $-\frac{1}{2}\log|\Sigma_k|$ — high-variance classes pay a spread penalty. New geometry: boundaries are conic sections (ellipses, parabolas, hyperbolas). Fit: same counting as LDA but $\hat{\Sigma}_k$ per class — needs enough samples in *every* class.

The island picture: one tight class, one loose → the boundary closes around the tight class. Far from both means, the wide Gaussian is the better explanation, so it wins everywhere outside. LDA on the same data draws a straight line — systematically wrong on one flank. That flank is LDA's bias, visible on a plot.

6 · LDA or QDA? Count your parameters

	LDA	QDA
Covariance parameters	p(p+1)/2 — shared	K · p(p+1)/2 — per class
p = 50, K = 3	1,275	3,825
Boundary family	hyperplanes	conics (curved)
Bias	high if spreads truly differ	low — flexible
Variance	low — pooled estimate	high — per-class estimates
Needs	modest n overall	large n in every class

Choose LDA when n is small, classes are scarce, or you want a stable, interpretable baseline. **Choose QDA** when n per class is generous and the spreads visibly differ. Middle ground: regularised DA — shrink each $\hat{\Sigma}_k$ toward the pooled $\hat{\Sigma}$ (Friedman 1989).

7 · LDA vs logistic regression

$$\log \frac{P(y=1|x)}{P(y=0|x)} = \beta_0 + \beta^\top x$$

Under LDA's assumptions the posterior log-odds are exactly linear — logistic regression's form. Same boundary family, different fit: LDA fits the joint P(x,y), using Gaussianity as extra statistical muscle (≈30% efficiency when true — Efron 1975) and staying stable when classes are nearly separable, but it is sensitive to outliers. Logistic fits the conditional P(y|x): no assumptions on x, robust to heavy tails, degrades gracefully — but its MLE blows up under perfect separation. Rule of thumb: ample data and prediction only → logistic; small n, K classes, or need densities → LDA.

8 · Fisher's view: the most separating direction

$$J(w) = \frac{w^\top S_B w}{w^\top S_W w}, \qquad w^* \propto S_W^{-1}(\mu_1 - \mu_2)$$

Maximise between-class scatter $S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^\top$ over within-class scatter S_W. The optimum is **not** the mean-difference line: S_W^{-1} tilts w to cancel the shared correlation — that tilt is the whole trick. Same w as the Gaussian/Bayes route: two derivations, one answer. After projection, classification collapses to a 1-D threshold; with K classes there are up to K−1 useful directions — a supervised map to K−1 dimensions.

	PCA	LDA (FISHER)
Uses labels?	no — unsupervised	yes — supervised
Maximises	total variance	between / within scatter
Directions	up to p	up to K − 1
Can bury class structure?	yes — variance ≠ separation	no — separation is the objective

> The classic pipeline: PCA first (compress, fix p > n singularity), LDA second (separate), threshold third — e.g. Fisherfaces.

9 · When discriminant analysis lies to you

- **Non-Gaussian classes:** skewed or multimodal classes put the boundary somewhere absurd. Fix: transforms, mixture DA, kernels.
- **Outliers:** $\hat{\mu}$ and $\hat{\Sigma}$ are not robust — one wild point tilts the boundary. Fix: robust estimates, trimming.
- **High dimensions:** $\hat{\Sigma}$ is singular when p ≥ n. Fix: shrinkage (Ledoit–Wolf), diagonal LDA, PCA first.
- **Imbalance and wrong priors:** log π_k is the principled lever — the sin is silently assuming 50/50 on skewed data.

> Habit: before trusting QDA's curves, look at per-class histograms / QQ plots. The model is checkable — so check it.

10 · Ninety years old and still shipping

- **Fisherfaces:** PCA + LDA on pixel vectors ran face recognition for a decade before CNNs.
- **Genomics (p ≫ n):** regularised/diagonal LDA classifies tumours from microarrays with n in the dozens.
- **Credit & finance:** Altman's Z-score (1968) — a linear discriminant on 5 ratios — still screens bankruptcy risk.
- **Brain–computer interfaces:** shrinkage LDA is the default EEG decoder: tiny n, real-time, stable.

The shared niche: small data, interpretability or latency constraints — where closed form beats gradient descent.

11 · The lecture in six lines

1. Classify to the biggest posterior: argmax π_k f_k(x) — the Bayes rule is the gold standard.
2. Model each class as a Gaussian — log-likelihoods become quadratics; algebra replaces integration.
3. Shared Σ → quadratics cancel → linear boundaries: LDA, fit by counting and averaging.
4. Per-class Σ_k → quadratics survive → conic boundaries: QDA — pay for flexibility with variance.
5. LDA vs logistic: same boundary family, generative vs conditional fit — efficiency vs robustness.
6. Fisher: maximise between/within scatter → same w, plus a (K−1)-dimensional supervised embedding.

12 · Going deeper

- James, Witten, Hastie, Tibshirani, *An Introduction to Statistical Learning*, ch. 4 — free PDF.
- Hastie, Tibshirani, Friedman, *The Elements of Statistical Learning*, §4.3 — LDA, QDA, logistic side by side.
- Bishop, *Pattern Recognition and Machine Learning*, §4.1–4.3 — the generative view.
- Fisher (1936), Annals of Eugenics 7. · Friedman (1989), JASA 84. · scikit-learn: LinearDiscriminantAnalysis (shrinkage included).